

Zhou Jefan

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Educational background

Wuhan University, Spatial Information and Digital Technology

, 2022.09-2026.06

- **GPA:** 3.74/4.0
- **Major courses:** C++ programming, University Physics, Discrete Mathematics, Data Structure and Algorithm, Operating System, Database System, Software Engineering, Frontier Technology of Artificial Intelligence, Computer Composition and Design, Principles of Computer Graphics, Human-Computer Interaction and Design Thinking, Spatio-Temporal Data Analysis and Mining, Software Architecture and Design Patterns, Java network programming, Digital image processing

Scientific Research experience

2024 College Students' Innovation and Entrepreneurship Program (National level), Group leader

2023.06-2024.06

Job content: Using Youtube online video data as the initial data. Then, the collected data were labeled according to the format, including the position of the vehicle bounding box and the category label.

Model selection and principle:

- **CountingMOT Object Tracking model:** Based on multi-object tracking technology, it can accurately count the number of vehicles by detecting and tracking them in a video sequence.
- **YOLOv7 object detection model:** It adopts the architecture of You Only Look Once, and realizes the rapid detection and location of multiple objects in the image through a single forward propagation.

Training process: Anaconda was used to configure PyTorch deep learning framework, and the model was trained in Wu Da Supercomputer Center. The vehicle data set was proportionally divided into training set, validation set and test set. The model is iteratively trained on the training set, the parameters are optimized to minimize the loss function, and the model is tuned on the validation set. The learning rate is dynamically adjusted by the optimizer and the learning rate scheduler to improve the training efficiency and model performance.

Training result metrics: For the CountingMOT model, the evaluation metrics include the accuracy of object tracking, the miss rate, the false detection rate, and the accuracy of counting the number of vehicles. For YOLOv7 model, the evaluation metrics include precision, recall, IoU, and mAP of object detection. Through the evaluation and comparison of the training results, we verify the effectiveness of the vehicle data set, and the performance of the model in vehicle detection and tracking tasks is excellent.

Research achievements: Second author of the paper "CSVD: A Cross-Scenario Vehicle Dataset for Multi-Object Tracking", published in SPIE International Conference proceedings, Scopus journal.

Project experience

Development of national nature reserve management system springboot API+vue, npm, Restful style

- The sql file was imported into the specified table, and the database was postgresSQL.

- Using IntelliJ Idea, through springboot architecture, develop apis related to adding, deleting, changing, and searching points (following a Restful style).
- VS Code was used to develop the front-end based on the vue architecture, and the back-end API was called in the front-end to execute the mouse click to realize the corresponding operation.

Raster data visualization service Java Servlet Java Jsp JDBC

- In accordance with OOP, In the Servlet section write Point.java, Line.java, Facet.java info.java as WKT points and lines in the Main function to read and import the three known Shape files into the local PostgreSQL.
- Call the Openlayer url in the Jsp page, and write the reading logic to read data from PostgreSQL, and the other part of the style design;

Red Army Long March route visualization static page Java jsp AmAP API IntelliJ IDEA

- Call the AmAP url, key key in jsp to display the map of China.
- According to the official API of AmAP and the known coordinates of Long March, mark each location and connect the adjacent locations with a straight line.
- Create a description for each location using css styles and tomcat publishing service.

Parallel client login, chat service design Java Servlet JavaFX MySQL IJ IDEA

Front-end (JavaFX) :

- **Design login interface:** Create login interface with JavaFX: user name, password input box, login button and other components, use layout manager for layout.
- **Design chat interface:** Create chat interface, including text area to display chat records, text box to input messages, send button and other components, and use layout manager to layout.
- **Implement event handling:** Add event handlers for components such as login button and send button, send login request to the backend when the login button is clicked, and send message to the backend when the send button is clicked.
- **Communicating with the backend:** Use the HTTP client library in JavaFX to send HTTP requests to the backend, including login requests and message sending requests, and process the response from the backend.

Backend (Java Servlet) :

- **Create a Servlet:** Create a Java Servlet class that handles HTTP requests from the client, including login requests and message sending requests.
- **Handle the login request:** handle the login request from the login interface in the Servlet, verify the user's identity, if the verification passes, create a user session, and use MySQL to store the user's password and account.
- **Handle the message sending request:** Handle the message sending request from the chat interface and broadcast the message to all online users.
- **Parallel processing:** Thread pool technology is used to realize parallel processing to ensure that requests from multiple clients can be processed at the same time.

Twitch+ : Personalized Twitch resource recommendation engine

- Designed and built a full-stack Spring Boot application that allows users to search Twitch resources (live streams/videos/clips) and get recommendations.
- A web page with a rich and user-friendly experience was built using React and Ant Design.
- MySQL database hosted on AWS RDS with Spring Data JDBC for CRUD operations.
- A RESTful controller API was implemented, and the Twitch API and OpenFeign HTTP client were used to retrieve real Twitch resources.
- Support for register/login/logout functionality implemented using Spring Security.
- Game information is extracted from Twitch resources, and a content-based recommendation algorithm is implemented based on favorites.
- We used Caffeine to implement Spring Boot caching and write unit tests.
- Containerized application deployment to AWS App Runner.

Next AI: A Web-based Full-stack Question-answering AI Agent for Querying PDF Documents

- An interactive session UI was created, leveraging React and Ant Design, to enable users to easily upload and interact with PDF documents in real time.
- A RESTful API was implemented with Express and Node.js, optimized for high performance request handling.
- Memory vectors are used to store cache-generated embeddings for efficient retrieval.
- OpenAI's GPT-3.5 Turbo API and Langchain technology are integrated to develop an advanced AI agent capable of loading, splitting, storing, retrieving, and outputting document functions.

Skills/Hobbies

- Language: English (CET-4: 574; Toefl: 104/120), proficient in listening, speaking and reading
- Programming language skills: C/C++, Java, Python, JavaScript/TypeScript, OpenGL, Matlab
- Language Framework: Spring Boot, Vue, Servlet/jsp, MVC, Node.js
- Development tools: Visual Studio, IntelliJ IDEA, PyCharm, VSCode, Angular